

Problem

A quantity that contains all the power information in a given load is the

- (a) power factor
- (b) apparent power
- (c) average power
- (d) reactive power
- (e) complex power

Step-by-step solution

Step 1 of 5

The value of power factor contains cosine of the angle between voltage and current. It doesn't contain any information about power. Hence, option (a) is incorrect.

Step 2 of 5

The apparent power is the product of the rms values of voltage and current. It contains power only in magnitude form. Hence, option (b) is incorrect.

Step 3 of 5

Average power is the instantaneous power over one period. Hence, option (c) is incorrect.

Step 4 of 5

Reactive power is contains power absorbed by only energy storing elements. Hence, it doesn't contain information about all power. Hence, option (d) is incorrect.

Step 5 of 5

Complex power is the product of rms voltage phasor and complex conjugate of the rms current phasor. It contains information about real and reactive power. Hence, **option (e)** is correct.